

VIII. APPENDIX

Theorem 1 (Privacy-Fairness-Utility Tradeoff of FedPF) Let $\hat{Y}$ be the output of the FedPF algorithm, and Y^* be the optimal classifier satisfying (ε_p, δ) -differential privacy (DP) and ε_f -fairness constraints (EO, DemP). If client data are independently and identically distributed and the Rademacher complexity of the classifier family is bounded, with probability at least $1 - \beta$, the following inequality holds:

$$\begin{aligned} \text{err}(\hat{Y}) &\leq \text{err}(Y^*) + O\left(\frac{B^2 \varepsilon_f^4 T^{3/2} \mathcal{H}}{\varepsilon_p^2}\right) + O\left(\mathfrak{R}_m(\mathcal{F}) + \sqrt{\frac{\log(1/\beta)}{m}}\right), \\ \text{err}(\hat{Y}) &\geq \text{err}(Y^*) + \Omega\left(\frac{|\mathcal{A}| \log(1/\varepsilon_f)}{\sqrt{m}} - \mathfrak{R}_m(\mathcal{F})\right) + \Omega\left(\sqrt{\frac{\log(1/\beta)}{m}}\right), \end{aligned} \quad (18)$$

where $\mathcal{H} = \ln(1/\delta) \ln^2(8T|\mathcal{A}|/\delta) \log(|\mathcal{K}| + 1)$. $\mathfrak{R}_m(\mathcal{F})$ is Rademacher complexity of the classifier family $\mathcal{F}$, $\mathfrak{R}_m(\mathcal{F}) \leq Cm^{-\alpha}$, $\alpha \leq 1/2$. B is the ℓ_1 -norm bound of dual variables λ , T is the number of training rounds. m is the client-side sample size. $|\mathcal{A}|$ is the number of sensitive attribute categories. $\mathcal{K}$ is a set of indices, and each index $k \in \mathcal{K}$ corresponds to a fairness constraint. If fairness constraints is DemP, then $|\mathcal{K}| = 2|\mathcal{A}|$, if fairness constraints is EO, then $|\mathcal{K}| = 4|\mathcal{A}|$. δ is the parameter of DP mechanism.

Proof (Sketch) The FedPF algorithm formulates the optimization as a zero-sum game between a learner (minimizing loss) and an auditor (enforcing fairness) under DP constraints. We derive the error bound as follows:

For the **upper bound**, the suboptimality gap $\text{err}(\hat{Y}) - \text{err}(Y^*)$ can be decomposed into Learner Regret and Auditor Regret from work [26]. However, The influence of the "fairness overfitting" phenomenon for model error cannot be ignored [5]. Thus, the error bound is considered Learner Regret (e_{LR}) and Auditor Regret (e_{AR}) and Generalization Error (e_G), as follows:

$$\begin{aligned} \text{err}(Y) - \text{err}(Y^*) &\leq \left(\frac{1}{T} \sum_{t=1}^T \mathcal{L}(f, \lambda) - \frac{1}{T} \min_{f \in \mathcal{F}} \sum_{t=1}^T \mathcal{L}(f, \lambda)\right) (e_{LR}) \\ &\quad + \left(\frac{1}{T} \max_{\lambda \in \Lambda} \sum_{t=1}^T \mathcal{L}(f, \lambda) - \frac{1}{T} \sum_{t=1}^T \mathcal{L}(f, \lambda)\right) (e_{AR}) \\ &\quad + [\text{err}(f) - \text{err}(f^*)] (e_G) \end{aligned} \quad (19)$$

1). **Learner Regret:** The learner minimizes a Lagrangian $\mathcal{L}(f, \lambda) = \mathbb{E}[\ell(f(x), y)] + \lambda \cdot \text{disc}(f)$, where λ are dual variables for fairness constraints. Using DP (Exponential Mechanism definition 5), noise scaled by ε_p is added to sensitive dataset. We can state the learner regret with probability at least $1 - \beta/2$ (in Lemma 4.1 [26]):

$$e_{LR} = O\left(\frac{B^2 \varepsilon_f^2 \sqrt{T \ln(1/\delta) \ln m}}{\varepsilon_p}\right). \quad (20)$$

2). **Auditor Regret:** The auditor updates λ to enforce $\text{disc}(f) \leq \varepsilon_f$. We can state the auditor regret with probability at least $1 - \beta/2$ (in Lemma 4.2 [26]):

$$e_{AR} = O\left(\frac{B^2 \varepsilon_f^4 T^{3/2} \ln(1/\delta) \ln^2(8T|\mathcal{A}|/\delta) \log(|\mathcal{K}| + 1)}{\varepsilon_p^2}\right). \quad (21)$$

3). **Generalization Error:** Using Rademacher complexity and standard concentration bounds [25], the generalization error under fairness constraint (EO and DemP) is:

$$e_G = O\left(\mathfrak{R}_m(\mathcal{F}) + \sqrt{\frac{\log(1/\beta)}{m}}\right). \quad (22)$$

Summing these terms yields the error bound. It can be seen from (20) and (21) that the impact of Auditor Regret is greater than that of Learner Regret.

For the **lower bound**, Fairness requires estimating group statistics $\gamma_{y,a}(f)$ (Definition 3) with precision ε_f . By Hoeffding's inequality [25], estimating differences $|\gamma_{y,a} - \gamma_{y,a'}|$ over $|\mathcal{A}|$ groups requires $\Omega(|\mathcal{A}|/\varepsilon_f^2)$ samples per group for variance control. With total m samples, the minimal detection error induces loss $\Omega(|\mathcal{A}|/(m\varepsilon_f^2))$. Taking log for entropy, we get $\Omega(|\mathcal{A}| \log(1/\varepsilon_f)/\sqrt{m})$ as the minimal utility penalty for strict ε_f . Moderate ε_f acts as regularization, reducing overfitting [5]. This subtracts from loss via Rademacher bound: gain $\leq -\mathfrak{R}_m(\mathcal{F})$ (negative, as fairness constraints prune complex discriminators). For strict ε_f , the penalty dominates, making the bound positive. Apply McDiarmid's inequality for concentration: the bound holds w.p. $1 - \beta$ with additive $\Omega(\sqrt{\log(1/\beta)/m})$ [25; 27]. Sum terms, yielding the lower bound. This matches the paper's non-monotonicity (bound dips then rises as $\varepsilon_f \downarrow$) and implies impossibility: no algorithm achieves $\Delta \text{err} < \Omega(\log(1/\varepsilon_f)/\sqrt{m})$ for small ε_f .

Theorem 2 (Extended for Non-I.I.D. of Theorem 1) Let $\hat{Y}$ be the output of the FedPF algorithm, and Y^* be the optimal classifier satisfying (ε_p, δ) -differential privacy (DP) and ε_f -fairness constraints (EO, DemP). Assume non-i.i.d. data with bounded heterogeneity $\Gamma = \max_{i,j} \|\nabla \mathcal{L}(D_i) - \nabla \mathcal{L}(D_j)\|^2$ (gradient dissimilarity across clients) and bounded local variance $\sigma^2 = \mathbb{E}_i[\text{Var}(\nabla \mathcal{L}(D_i))]$. If the Rademacher complexity of the classifier family is bounded, with probability at least $1 - \beta$, the following inequalities hold:

$$\begin{aligned} \text{err}(\hat{Y}) &\leq \text{err}(Y^*) + O(\mathcal{A}) + O\left(\mathcal{B} + \frac{\Gamma T}{N} + \sigma^2 \sqrt{\frac{T}{mN}}\right), \\ \text{err}(\hat{Y}) &\geq \text{err}(Y^*) + \Omega\left(\mathcal{C} + \frac{\Gamma T}{N} + \sigma^2 \sqrt{\frac{T}{mN}}\right) + \Omega\left(\sqrt{\frac{\log(1/\beta)}{m}}\right), \end{aligned} \quad (23)$$

where all terms match Theorem 1, $\mathcal{A} = \frac{B^2 \varepsilon_f^4 T^{3/2} \mathcal{H}}{\varepsilon_p^2}$, $\mathcal{B} = \mathfrak{R}_m(\mathcal{F}) + \sqrt{\frac{\log(1/\beta)}{m}}$, $\mathcal{C} = \frac{|\mathcal{A}| \log(1/\varepsilon_f)}{\sqrt{m}} - \mathfrak{R}_m(\mathcal{F})$, plus non-i.i.d. penalties: $\frac{\Gamma T}{N}$ (heterogeneity drift over T rounds, averaged over N clients) and $\sigma^2 \sqrt{\frac{T}{mN}}$ (local variance amplified by heterogeneity). This extends Theorem 1 by incorporating data heterogeneity, showing that non-i.i.d. distributions inflate both upper and lower bounds on error, exacerbating privacy-fairness-utility tradeoffs.

Proof (Sketch) The proof extends Theorem 1's decomposition ($e_{LR} + e_{AR} + e_G$) to non-i.i.d. settings, incorporating heterogeneity measures from FL convergence analyses.

1. **Learner Regret (e_{LR}):** In non-i.i.d., local updates drift due to heterogeneous gradients. Extend Eq. (20) with $\frac{\Gamma T}{N}$ (drift penalty) [1]: $e_{LR} = O\left(\frac{B^2 \mathcal{A}^2 \sqrt{T \ln(1/\delta) \ln m}}{\varepsilon_p} + \frac{\Gamma T}{N}\right)$. For lower bound, heterogeneity adds minimal drift $\Omega(\Gamma T/N)$.

2. **Auditor Regret (e_{AR}):** Fairness enforcement via λ updates is distorted by heterogeneous attribute distributions. Extend Eq. (21) with variance term: $e_{AR} = O\left(\frac{B^2 \varepsilon_f^4 T^{3/2} \ln(1/\delta) \ln^2(8T|\mathcal{A}|/\delta) \log(|\mathcal{K}| + 1)}{\varepsilon_p^2} + \sigma^2 \sqrt{\frac{T}{mN}}\right)$. Lower bound mirrors with $\Omega(\sigma^2 \sqrt{T/(mN)})$.

3. **Generalization Error (e_G):** Retain Rademacher-based bound from Eq. (22), but non-i.i.d. increases effective complexity via variance, lower bound follows similarly with added heterogeneity terms.

Summing yields the bounds. The non-i.i.d. terms inflate error (upper/lower), implying worse tradeoffs, e.g., stricter ε_f or smaller

ε_p amplifies drift, requiring more clients N or rounds T to mitigate. Concentration holds w.p. $1-\beta$ via McDiarmid's inequality [25; 27].

A. Convergence and Robustness

We analyze robustness to distribution shifts between sensitive groups (p) and protected groups ($\hat{p}$) using the Total Variation (TV) distance, i.e. $TV(p, \hat{p})$, which measures distribution divergence between p and $\hat{p}$.

Theorem 3 (Fairness Discrimination Bound of FedPF) *The total fairness discrimination across all clients is bounded by:*

$$\sum_{i=1}^{\mathcal{N}} (\gamma_{y, \hat{p}, i}(f_i) - \gamma_{y, p, i}(f_i)) \leq \mathcal{N} \cdot \alpha_{\max}, \quad (24)$$

where $\alpha_{\max} = \max_{i \in \mathcal{N}} \{TV(p_i, \hat{p}_i)\}$.

Proof (Sketch) For any group label $a \sim p$, $a' \sim \hat{p}$ of client i , from Theorem 1 in work [30], we have:

$$\begin{aligned} \sum_{i=1}^{\mathcal{N}} \gamma_{y, a}(f_i) &= \sum_{i=1}^{\mathcal{N}} \{\gamma_{y, a}(f_i) - \gamma_{y, a'}(f_i) + \gamma_{y, a'}(f_i)\}, \\ &\leq \sum_{i=1}^{\mathcal{N}} \{|\gamma_{y, a}(f_i) - \gamma_{y, a'}(f_i)| + \gamma_{y, a'}(f_i)\}. \end{aligned} \quad (25)$$

From the Kantorovich-Rubenstein [31], we obtain:

$$|\gamma_{y, a}(f) - \gamma_{y, a'}(f)| = |\mathbb{E}_{a \sim p}[f(\theta)] - \mathbb{E}_{a' \sim \hat{p}}[f(\theta)]| \leq TV(p_j, \hat{p}_j). \quad (26)$$

Therefore, $\sum_{i=1}^{\mathcal{N}} |\gamma_{ij}(f_i) - \hat{\gamma}_{ij}(f_i)|$ has a deterministic upper bound if $TV(p_i, \hat{p}_i) \leq \alpha_i$ for each client $i \in \mathcal{N}$, where the parameter α_i is a constant. Assume that the parameter $\max_{i \in \mathcal{N}} \alpha_i = \alpha_{\max}$, then we have the upper bound of the distance $\sum_{i=1}^{\mathcal{N}} (\mathcal{G}_{ij}(f_i) - \hat{\mathcal{G}}_{ij}(f_i))$, as follows:

$$\sum_{i=1}^{\mathcal{N}} (\gamma_{y, \hat{p}, i}(f_i) - \gamma_{y, p, i}(f_i)) \leq \sum_{i=1}^{\mathcal{N}} TV(p_i, \hat{p}_i) \leq \mathcal{N} \cdot \alpha_{\max}. \quad (27)$$

where $\alpha_{\max} = \max_{i \in \mathcal{N}} \{TV(p_i, \hat{p}_i)\}$.